Backdoor Defense via Malicious Knowledge Capturing and Machine Unlearning with Out-of-Distribution Data

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***Abstract*—Recent studies reveal that backdoor attacks are a significant security threat to deep neural networks. Existing defense methods for removing backdoors from victim models often assume that defenders have access to limited clean training data, which is unrealistic in real-world scenarios. Inspired by the biological immune response process, we propose a novel backdoor defense framework comprises three key steps. First, a max-entropy staircase approximator (MSA) is implemented to capture the valid trigger distribution through trigger reconstruction. We then utilize these reconstructed trigger patterns to extract backdoor knowledge from the victim model using a knowledge distillation-based approach. Finally, to completely eliminate the backdoor knowledge, we design a machine unlearning method optimized by maximizing the KL-divergence between the backdoor knowledge representation and the victim model. Our framework relies solely on unlabeled out-of-distribution data, significantly enhancing its practicality. Extensive experiments on various backdoor attacks show that our approach can lower the attack success rates of five state-of-the-art backdoor attacks by an average of 99% and outperforms five state-of-the-art defense methods that rely on clean training data.**

***Index Terms*—Backdoor Defense, Trigger Pattern Recovery, Machine Unlearning**

I. Introduction

Despite the impressive performance of deep neural networks (DNNs) in fields such as facial recognition and autonomous driving, numerous studies have shown that they are vulnerable to backdoor attacks [1, 2]. In backdoor attacks, an adversary can poison the training dataset [3, 4] or tamper with model parameters during the training process [5, 6], causing the model to behave normally on clean samples but misclassify samples containing backdoor triggers to the target label predefined by the adversary. Since training DNNs requires large datasets and significant computational resources, companies or individuals often download datasets from the internet for training, rely on untrusted third-party platforms for model training, or use pre-trained models from untrusted providers. These are common scenarios where backdoor attacks can occur, posing a significant threat to the deployment and application of DNNs, and underscoring the importance of backdoor defense.

To counter backdoor attacks, many defense methods have been developed, which can be broadly categorized into two types. **1)** Training-stage methods [7, 8, 9], which aim to train a benign model directly from a contaminated dataset using carefully designed techniques. **2)** Post-training methods [10, 11, 12], which purify the victim model by adjusting its internal parameters, while maintaining its prediction accuracy on benign samples. In this work, we focus on the latter. Since the neurons affected by backdoor attacks are entangled with normal neurons in the victim model, removing backdoors is highly challenging. Existing defense methods often assume that the defender has access to 1%-5% of clean training data, which is impractical in real-world scenarios. Moreover, adjusting all the model parameters is computationally costly.

To address the aforementioned challenges, we propose a novel and more practical method that requires no training data or labels, enabling defenders to perform backdoor defense simply by collecting some unlabeled out-of-distribution samples. Since both antigens and backdoor triggers can induce abnormal behavior, we develop our defense method inspired by the biological immune response, as shown in Fig. 1, which comprises three steps: first, the presentation of foreign antigens; second, the generation of specific antibodies; and finally, the immune-mediated elimination of invaders. When a biological organism detects an antigen, it stimulates specific cells to produce antibodies, thereby completing the clearance of foreign substances. Correspondingly, we first capture the valid trigger distribution via a trigger reconstruction method, then a knowledge distillation based approach is employed to obtain the backdoor knowledge representation, and finally we leverage it as a guidance to eliminate the backdoor behavior from the victim model.

Specifically, we first introduce the max-entropy staircase approximator (MSA) [13] to reconstruct the backdoor trigger. Studies show that knowledge distillation with clean samples encourages benign knowledge transfer from the teacher to the student model, yielding an initially purified model [14, 15]. Intuitively, we use backdoor-triggered samples during distillation process to promote the transfer of backdoor knowledge, resulting in a model that primarily contains most of the victim model's key backdoor knowledge, which we call the backdoor knowledge representation model. Finally, we design a machine unlearning method that maximizes KL-divergence between the

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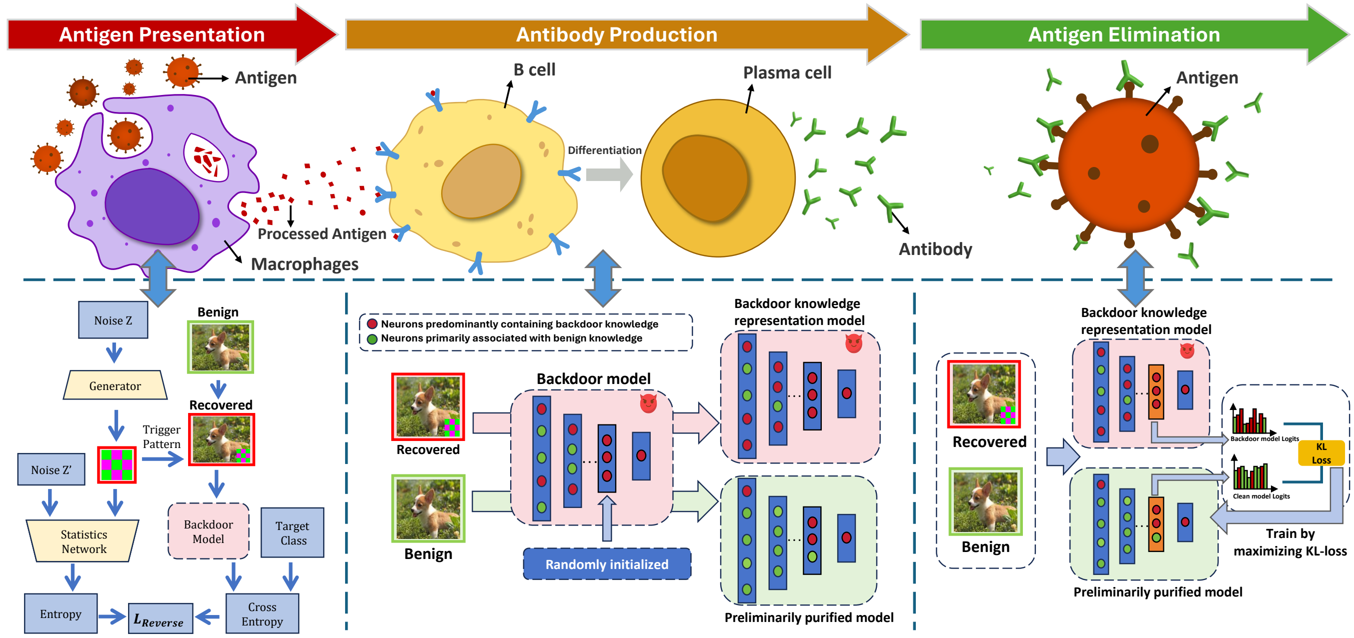


Fig. 1: **Top:** Illustration of the immune response process, primarily including antigen presentation, antibody production, and antigen elimination. **Bottom:** Our proposed backdoor defense framework inspired by the immune response process.

backdoor knowledge representation model and a preliminarily purified version of the victim model to completely eliminate the backdoor behavior from it. Our contributions are as follows:

1. Inspired by the immune response, we propose a novel framework for backdoor defense that allows defenders to utilize unlabeled out-of-distribution data, without the need for access to benign training samples. This enables users to perform backdoor defense using their own collected data, making the approach more practical in real-world scenarios.

2. We discover that during knowledge distillation, querying the teacher model with backdoor samples can promote the transfer of its backdoor knowledge to the student model, enabling us to extract the backdoor knowledge representation from the victim model.

3. We carefully design a KL-divergence maximization based machine unlearning method that can leverage the backdoor knowledge representation to effectively eliminate the backdoor without the dependency for extra clean samples.

4. Extensive experiments on standard image datasets demonstrate that our framework effectively reduces the attack success rates of five state-of-the-art backdoor attacks by 99% and achieves an average defense effectiveness rate of 96.44%. This performance surpasses a range of state-of-the-art defense methods that rely on labeled training samples.

II. Related Work

A. Backdoor Attack

In traditional backdoor attacks, the adversary modifies a subset of the training dataset by embedding a specific pattern and assigning these modified samples to the target label. A backdoor model trained on such poisoned dataset which contains manipulated samples and benign ones will correctly classify clean samples while predict backdoor samples as the target label during the inference stage.

Various kinds of triggers have been shown to be capable of redirecting the predictions of compromised models, with the simplest and most common being a black-and-white patch or a blending pattern [16, 17]. Although these patterns are effective and not subtle, they can still be detected by human eyes. In recent years, more complex and stealthy backdoor triggers [6, 18, 19] have been developed. Refool generates triggers by using mathematical modeling of physical reflection models. WaNet introduces a novel training approach called the "noise mode" to design triggers. Ftrojan implements backdoor attacks in a black-box setting by trojaning the frequency domain of images. Besides, clean-label attacks [20, 21, 22] further increase the stealthiness of backdoor attacks by obfuscating the main content of the image and establishing a correlation between the trigger and the target label without altering the poisoned sample's label.

B. Backdoor Defense

Backdoor defense methods can mainly be divided into two categories: Training-stage defenses and Post-processing defenses. In Training-stage defenses, the defender's goal is to train a clean model from a poisoned dataset directly. More backdoor defense efforts are classified as Post-processing defenses, where the objective is to remove the backdoor from a backdoored model while maintaining its accuracy on clean samples.

a) Trigger reconstruction defense: A significant portion of these works implements backdoor defenses by first reconstructing the backdoor trigger through reverse engineering, then fine-tuning the model using a subset of clean training samples [23, 24]. To the best of our knowledge, Neural Cleanse (NC) proposed the first trigger synthesis-based defense, where defenders first obtained potential trigger patterns for each class, and then used Median Absolute Deviation (MAD) for anomaly detection to determine the final synthetic trigger and its target label. ABS introduces a new method to study neuron behavior by observing how output activations vary with different levels of neuron stimulation. To further improve the quality of trigger reconstruction, methods [25, 26] have been proposed, often relying on classic generative models like GANs or VAEs.

b) Pruning based defense: Another major category of backdoor defense methods is based on neuron pruning techniques. Fine-Pruning (FP) [10] proposes a method that prunes neurons highly correlated with backdoor behavior and then fine-tunes the pruned network. Adversarial Neuron Pruning (ANP) [11] designs an algorithm to prune neurons sensitive to adversarial neuron perturbations. Zheng et al. [27] locates backdoor-sensitive neurons by distinguishing their pre-activations on a poisoned dataset.

c) Knowledge distillation based defense: Knowledge distillation has been introduced to backdoor defense. Neural Attention Distillation (NAD) [28] proposes a backdoor defense method that integrates fine-tuning, knowledge distillation, and neural attention transfer. Building on NAD, Attention Relation Graph Distillation (ARGD) [29] introduces an improved backdoor defense framework that fully considers the correlation of attention features of different orders. Recently, Backdoor Cleansing with Unlabeled Data (BCU) [15] employs knowledge distillation and a carefully designed layer-wise weight reinitialization strategy for backdoor defense under unlabeled data conditions.

Although most of these methods can eliminate backdoor effectively from the victim model, they share a common assumption that defenders have access to clean training samples, which is often impractical in real-world scenarios. In contrast, our method uses unlabeled out-of-distribution samples for defense, requiring no access to training data or labels.

III. Method

A. Problem Formulation

Defender's goal. We assume that the defender has downloaded the backdoored model f_t from an untrusted online platform and does not have access to the original training process or any of the training data. The defender's objective is to derive a new model $\hat{f}$ using f_t and a limited set of out-of-distribution clean images, denoted as $D_{\text{out}}\{(x_i, y_i)\}_{i=1}^{M}$. The aim is to preserve high classification accuracy (ACC) on clean data while simultaneously reducing the attack success rate (ASR) on backdoored images.

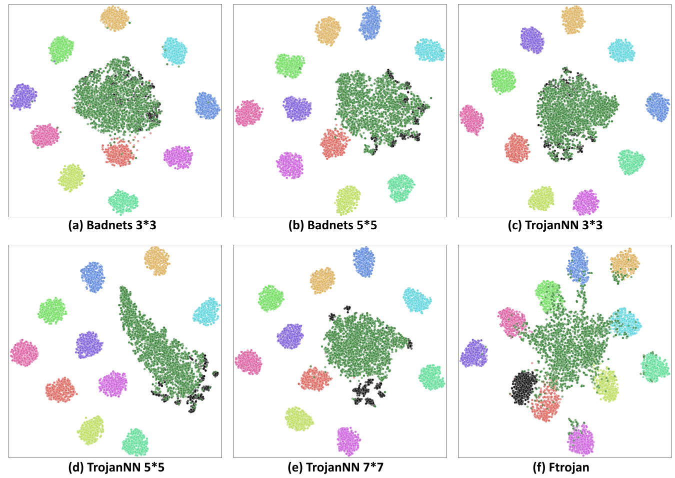


Fig. 2: T-SNE visualization of penultimate features on CIFAR-10 from various attacks. **Black**: Original backdoor samples. **Dark green**: Out-of-distribution imgaes with triggers recovered by MSA.

B. Capturing valid trigger distribution via trigger reconstruction

Previous study [13] has shown that backdoored models generalize their original triggers during backdoor injection, suggesting that the valid trigger distribution should be broader than the original trigger distribution. In this section, rather than attempting to design triggers identical to the original, which is impractical for implicit or complex trigger patterns, we aim to employ trigger reconstruction method to generate triggers that approximate the original ones at the feature level and could induce the same backdoor behavior in the model.

Given that the defender lacks knowledge of the precise distribution of the backdoor trigger, we approach this challenge as a sampling-free generative modeling problem. Our objective is to design a generative model G, which takes an n-dimensional noise vector Z as input, generating an output $G(Z)$ that approximates σ, which is the perturbation added to the sample and causes misclassification in the model.

However, typical generative models, such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), require sampling from the true data distribution and often suffer from mode collapse and convergence instability during training. Therefore, we introduce the Max-Entropy Staircase Approximator (MSA) to address this issue [13, 30]. Fig. 1 illustrates the overall process of the MSA method. Specifically, this method estimates the trigger distribution σ using multiple sub-models $G = \{G_1, \ldots, G_n\}$, with each sub-model learning a portion of σ based on staircase approximation. By setting multiple thresholds $\gamma_{1,\ldots,N} \in [0, 1]$, the domain of the function F is divided into multiple intervals (here $\gamma_{i+1} > \gamma_i$). For each threshold γ_i, a sub-interval $X' = \{x' : F(x') > \gamma_i\}$ is defined, where γ_i densely covers the range [0,1]. This allows each G_i to capture behavior within a specific interval, and the sub-models are integrated to form the final generative model G. During training, the generative model G_i is optimized using the following loss function:

$$L_{\text{Reverse}} = \frac{1}{b} \sum_{x \in D_c} (\max(0, \gamma_i - F(x + G_i(z))) \\ - \beta I_i(G_i(z); z')) \tag{1}$$

where b is the batch size, D_c is the set of benign samples, and β is a hyperparameter used to balance the regularization term and the mutual information term. I_i is the mutual information estimator that measures the dependence between $G_i(z)$ and z', where z and z' are randomly sampled noise from the interval [0,1].

Qualitative Analysis. In order to verify that the MSA method we introduced effectively captures the characteristics of the original trigger, we utilized the T-SNE [31] technique to visualize the feature representations of clean images, backdoor samples, and out-of-distribution (OOD) data with recovered triggers generated by MSA. As shown in Fig. 2, for patch-based backdoor attacks such as Badnets [16] and TrojanNN [32], the dimensionality-reduced samples with recovered triggers align closely with the true backdoor samples (depicted by black points), indicating consistency at the feature level. We can also observe that the distribution of the recovered samples is an extension of the original backdoor samples. This observation is largely consistent with the conclusion of study [13] that valid trigger distribution should be broader than the original one. Even for less visible trigger types, such as Ftrojan backdoor attack [19], some dimensionality-reduced recovered samples still overlap with the black points, indicating that the method effectively captures the features of the original trigger.

C. Backdoor Representation Extraction

Intuitively, backdoor samples induce abnormal behavior in a compromised model. To exploit this, we employ a knowledge distillation framework. Specifically, we query the teacher model using out-of-distribution (OOD) data embedded with reconstructed triggers, allowing the transfer of information from neurons strongly associated with backdoor behavior to the student model. The goal is to extract the backdoor knowledge representation from the backdoored model. Since we use the teacher model's logits as supervision, the true labels of the data samples are not required during the representation extraction process.

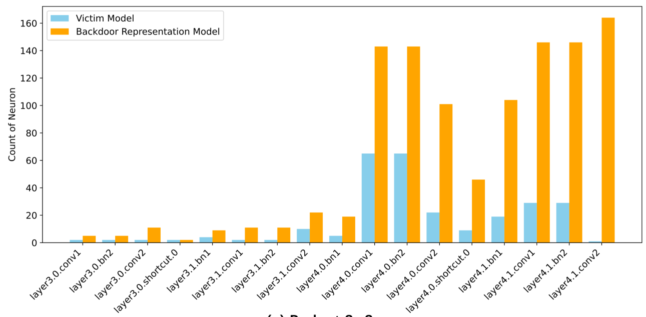
To enhance defense effectiveness and reduce backdoor removal time, we also query the teacher model with clean OOD samples to distillate benign knowledge to the student model while ignoring backdoor-related knowledge, producing a preliminarily purified model. Unlike direct knowledge distillation, we use adaptive layered weight initialization tactic [15], which randomly reinitializes certain neuron weights to remove backdoor knowledge. Prior research [33] indicates deeper layers are more associated with backdoor activation, and fine-tuning only the penultimate or antepenultimate layers can achieve similar results as fine-tuning all parameters. Therefore, we modify the random parameter initialization setup to apply only to the penultimate layer, instead of all layers, effectively disconnecting triggers from target labels while preserving accuracy on clean samples. This sets a solid foundation for subsequent backdoor removal. Detailed implementation of the Representation Extraction stage is provided in Algorithm 1.

Algorithm 1 Representation Extraction and Unlearning

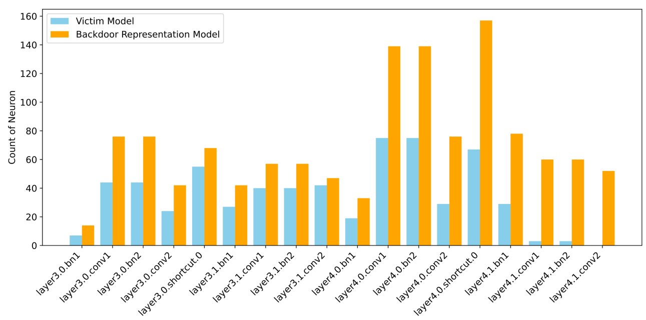
Input: Backdoor model f^t, random initialized student model f^s, model weights of the l-th layer W_l, initialization ratio δ, OOD data D_{defense}, samples with recovered trigger $D_{\text{recovered}}$, extraction epochs E_1, iterations I, unlearning epochs E_2.
Output: Clean model F_{clean}

 # Backdoor Representation Extraction
$Mask = \{m_l \mid m_l \in \{0,1\}^{\text{shape}(W_l^s)}, \sum m_l = \delta_l |m_l|\}$
Obtain weight mask m_l, $l = |W^t| - 1$.
$W_l^{\text{pre}} = (1 - m_l) \odot W_l^t + m_l \odot W_l^s$, $l = |W^t| - 1$
Initialize *phase* $\leftarrow$ "bad"
for $e = 0$ to E_1 **do**
    for $i = 0$ to I **do**
        Sample mini-batches b from $D_{\text{recovered}}$ if *phase* = "bad" else D_{defense}
        Obtain temperature-scaled probabilities from f^t and f^{phase}
        Update W^{phase} with Eq. 4
    end for
    $f^{\text{phase}'} \leftarrow f^{\text{phase}}$
    phase $\leftarrow$ "pre" if *phase* = "bad" else "bad"
end for
 # Backdoor Unlearning
Freeze $f^{\text{bad}'}$ and all layers of $f^{\text{pre}'}$ except the penultimate
$D_{\text{unlearning}} = \text{mix}(D_{\text{defense}}, D_{\text{recovered}}, 2:3)$
for $e = 0$ to E_2 **do**
    for $i = 0$ to I **do**
        Sample mini-batches b from $D_{\text{unlearning}}$
        Obtain temperature-scaled probability vectors from $f^{\text{pre}'}$ and $f^{\text{bad}'}$
        Update $W_l^{\text{pre}'}$ with Eq. 5, $l = |W^t| - 1$.
    end for
    $F_{\text{pre}} \leftarrow f^{\text{pre}'}$
end for

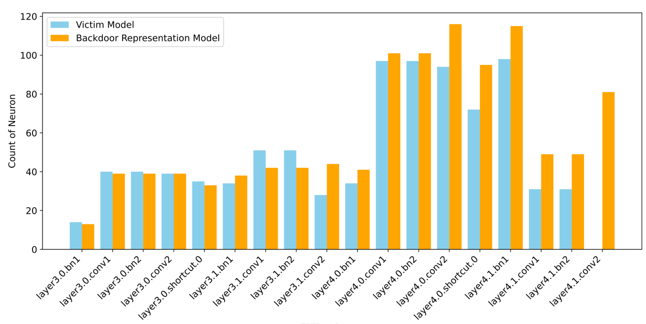
Qualitative Analysis. To validate the effectiveness of the backdoor representation extraction process, we visualize the activation distribution of both the original backdoored model and the backdoor representation model. Specifically, we calculate the top 500 images that most activate neurons in the penultimate and antepenultimate layers of the model and analyze the class distribution of these images. We quantify the number of neurons whose top-500 images contain more than 50% backdoor images. A higher count indicates that more neurons in the model are sensitive to backdoor samples, thus containing more backdoor-related information. As shown in Fig. 3, we present the experimental results of the explicit trigger-based



(a) Badnet 3×3



(b) TrojanNN 3×3



(c) FTrojan

Fig. 3: The activation distribution of both the original backdoored model and the backdoor representation model.

backdoor attacks BadNets [16] and Trojan backdoor attack [32], as well as the covert frequency-domain backdoor attack FTrojan [19]. It is observable that the backdoor representation model has more backdoor-sensitive neurons than the original backdoored model, and this phenomenon is more prominent in the penultimate layer. This also explains why we select this layer for unlearning during the backdoor unlearning phase.

D. Backdoor Cleansing via Machine Unlearning

Even though we query the teacher model with benign OOD data, this process inevitably transfers some backdoor knowledge to the student model. According to Algorithm 1, we design a backdoor unlearning based algorithm optimized by maximizing the KL-divergence between the backdoor knowledge representation and the pre-purified model to completely remove the remaining backdoor knowledge. Let h_1 and h_2 represent the output vectors from the penultimate layer of the pre-purified model and backdoor knowledge representation model, respectively. Their probability vectors can be expressed as:

$$P_{\text{pre}}(z) = \frac{\exp(h_1)}{\sum_k \exp(h_1[k])} \tag{2}$$

$$P_{\text{bad}}(z) = \frac{\exp(h_2)}{\sum_k \exp(h_2[k])} \tag{3}$$

where z represents the possible output classes. KL-divergence $D_{KL}(P \| Q)$ is defined as:

$$D_{KL}(P_{\text{pre}} \| P_{\text{bad}}) = \sum_z P_{\text{pre}}(z) \log \frac{P_{\text{pre}}(z)}{P_{\text{bad}}(z)} \tag{4}$$

As shown in Fig. 1, we feed both benign out-of-distribution data and samples with recovered triggers into the pre-purified model and the backdoor knowledge representation model. From each model, we extract the embedding vectors from the penultimate layer and calculate the KL-divergence. The pre-purified model is then optimized using the following objective function:

$$\begin{aligned} L_{\text{Defense}} &= -D_{KL}(P_{\text{pre}} \| P_{\text{bad}}) \\ &= -\sum_z P_{\text{pre}}(z) \log \frac{P_{\text{pre}}(z)}{P_{\text{bad}}(z)} \end{aligned} \tag{5}$$

Since the backdoor knowledge representation model captures most of the suspicious teacher model's malicious behavior along with a small portion of benign information, in contrast, the pre-purified model retains less backdoor content and a greater amount of clean knowledge. Intuitively, an increasing KL-divergence between this representation and the pre-purified model suggests that the latter is progressively shedding its backdoor traits while preserving a stable level of benign knowledge. It is worth noting that during this step, the backdoor knowledge representation model remains frozen. In this way, we can effectively and thoroughly eliminate backdoor knowledge from the model without significantly affecting its prediction performance on clean images.

IV. Experiments

A. Experiments settings

Datasets and Architecture. We conduct all backdoor models on two datasets include CIFAR-10 and CIFAR-100 [34]. The amount of clean data available to the defender is set to 2500. PreAct-ResNet18 [35] is adopted as the model architecture.

Backdoor attacks setting. We evaluated the effectiveness of the proposed method on five well-known backdoor attacks, including BadNets [16], Blended attack (Blended) [17], Trojan backdoor attack (TrojanNN)[32], Input-aware dynamic backdoor attack (IAB) [5], and Ftrojan [19]. Specifically, BadNets and TrojanNN are classic patch-based visible backdoor attacks. For BadNet and TrojanNN, we also consider the attack performance under different trigger sizes. Blended is a backdoor attack based on noise or semi-transparent image

TABLE I: Comparison with the state-of-the-art defenses on CIFAR-10 dataset on PreAct-ResNet18.(%)

ATTACK		Original			FT			FP			i-BAU			NAD			ANP		
Attack method	Trigger size	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑
Badnet	3×3	91.63	99.98	-	**91.64**	2.48	98.69	89.90	0.36	98.88	84.44	1.74	95.46	89.15	47.10	75.13	**91.64**	1.13	**99.36**
	5×5	93.41	100	-	**92.59**	19.35	89.39	90.69	4.09	96.07	86.62	2.20	94.98	92.46	54.73	71.64	91.15	1.53	97.58
	7×7	93.37	100	-	92.63	34.65	82.18	90.66	24.58	86.23	86.25	1.70	95.46	**92.64**	39.23	79.90	90.47	4.09	96.38
TrojanNN	3×3	91.87	99.75	-	**91.82**	1.70	97.52	89.50	0.55	97.07	85.48	1.97	94.35	89.00	40.54	76.83	87.50	14.19	89.25
	5×5	93.28	100	-	**92.73**	25.67	86.87	90.24	35.59	80.67	85.64	3.27	94.53	92.50	37.04	81.08	92.72	8.68	95.36
	7×7	93.43	100	-	92.64	46.29	76.46	90.59	12.42	92.37	86.19	2.08	95.34	92.40	30.39	84.29	**92.65**	15.69	91.77
Ftrojan	-	93.32	100	-	**92.70**	61.66	68.86	90.44	92.56	52.28	85.59	18.95	86.65	92.43	20.58	89.26	90.94	5.53	96.04
Blended	-	93.36	99.71	-	92.69	83.71	56.66	90.46	10.33	92.24	86.68	5.66	92.68	92.67	98.73	50.17	**92.90**	96.23	50.51
Input-aware	-	91.32	96.85	-	**93.51**	68.70	62.98	91.37	24.56	86.12	85.59	19.37	85.88	92.00	95.60	50.28	91.14	0.56	**98.06**
Avg	-	92.77	99.58	-	**92.43**	34.44	82.08	90.31	22.56	86.98	85.86	4.70	93.68	91.66	46.04	76.04	91.25	18.38	89.53

ATTACK		BCU(Tiny)			BCU(GTSRB)			BCU(CIFAR-100)			Ours(Tiny)			Ours(GTSRB)			Ours(CIFAR-100)		
Attack method	Trigger size	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑
Badnet	3×3	89.46	9.57	94.12	88.30	2.60	97.02	89.92	17.46	90.40	90.15	0.63	98.94	74.08	**0.34**	91.04	90.14	0.56	98.96
	5×5	90.61	2.45	97.38	89.80	1.64	97.38	90.35	4.03	96.46	90.46	0.13	98.46	79.82	**0.11**	93.15	91.16	0.13	**98.81**
	7x7	90.78	1.63	97.89	90.56	1.60	97.80	91.52	12.48	92.84	90.83	0.08	98.69	82.45	0.08	94.50	91.07	**0.07**	**98.81**
TrojanNN	3×3	91.26	29.88	84.63	84.44	22.32	85.00	91.23	34.06	82.52	90.28	**0.17**	99.00	81.86	0.37	94.68	90.49	0.20	**99.08**
	5×5	92.55	20.84	89.22	90.33	3.25	96.90	92.66	13.62	92.88	90.34	0.12	98.47	80.26	**0.06**	93.46	91.62	0.24	**99.05**
	7×7	92.23	30.13	84.34	90.29	10.08	93.39	**92.65**	17.24	90.99	91.31	0.16	**98.86**	82.09	**0.08**	94.29	91.13	0.09	98.80
Ftronjan	-	91.51	61.33	68.43	86.62	30.79	81.26	91.34	61.40	68.31	90.54	0.92	**98.15**	82.30	**0.31**	94.34	89.45	0.86	97.64
Blended	-	92.55	55.71	71.60	87.62	3.44	95.26	91.50	27.38	85.24	89.47	0.54	**97.64**	77.29	**0.03**	91.80	89.49	0.79	97.52
Input-aware	-	90.86	7.37	94.51	83.71	0.76	94.24	90.76	9.23	93.53	88.94	0.86	96.80	70.00	**0.39**	87.57	86.44	0.98	95.49
Avg	-	91.31	24.32	86.90	87.96	8.49	93.14	91.32	21.87	88.13	90.25	0.40	**98.33**	78.9	**0.19**	92.76	90.11	0.43	98.24

TABLE II: Comparison with the state-of-the-art defenses on CIFAR-100 dataset on PreAct-ResNet18.(%)

ATTACK		Original			FT			FP			i-BAU			NAD			ANP		
Attack method	Trigger size	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑
Badnet	3×3	67.98	99.63	-	65.22	1.54	**97.66**	25.83	42.93	57.27	28.56	8.58	75.81	**68.55**	55.68	71.69	66.36	10.71	93.65
	5×5	69.31	100	-	67.54	98.85	49.69	25.83	42.92	56.80	25.00	65.30	45.19	61.12	80.89	55.46	**67.69**	78.83	59.78
	7×7	69.90	100	-	**68.03**	100	49.06	25.18	72.31	41.48	25.25	61.49	46.93	62.45	100	46.28	67.80	98.45	49.72
TrojanNN	3×3	68.05	98.32	-	65.74	0.80	**97.60**	45.09	0.74	87.31	28.67	10.63	74.16	57.57	64.26	61.79	65.66	6.79	94.57
	5×5	70.25	100	-	68.01	97.83	49.96	26.36	45.55	55.28	25.35	66.16	44.47	63.89	88.79	52.42	**70.11**	99.83	50.02
	7×7	69.72	100	-	**68.05**	98.61	49.86	23.17	34.64	59.40	26.07	52.20	52.08	64.19	96.19	49.14	67.83	94.80	51.66
Ftrojan	-	69.61	99.77	-	**68.05**	3.88	97.16	24.67	59.56	47.64	24.63	52.89	50.95	62.07	20.08	86.08	67.90	0.23	**98.92**
Blended	-	69.91	99.57	-	67.23	99.24	48.82	23.88	32.41	60.57	25.63	39.26	58.01	60.48	98.50	45.82	**68.82**	97.70	50.39
Input-aware	-	63.78	94.30	-	**68.55**	97.90	49.42	42.84	98.95	41.86	36.74	76.47	45.40	63.06	95.98	50.48	61.99	18.43	87.04
Avg	-	68.98	98.45	-	**67.38**	66.52	65.16	29.21	47.78	55.45	27.32	48.11	54.34	62.60	77.82	57.12	67.13	56.20	70.20

ATTACK		BCU(Tiny)			BCU(GTSRB)			BCU(CIFAR-10)			Ours(Tiny)			Ours(GTSRB)			Ours(CIFAR-10)		
Attack method	Trigger size	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑	ACC↑	ASR↓	DER↑
Badnet	3×3	59.62	27.69	81.79	61.37	46.01	73.50	59.50	11.87	89.64	61.55	**0.21**	96.50	54.39	0.73	92.66	58.13	0.23	94.78
	5×5	59.35	11.73	89.16	61.23	0.09	**95.92**	60.36	6.02	92.51	59.03	0.35	94.68	58.84	**0.02**	94.76	59.34	0.07	94.98
	7×7	59.83	52.10	68.91	55.12	13.03	86.10	58.31	5.62	91.40	58.32	**0.23**	94.10	56.44	0.70	92.92	58.40	0.26	**94.12**
TrojanNN	3×3	**66.80**	87.13	54.97	60.56	47.61	71.61	66.04	79.12	58.60	62.27	**0.55**	96.00	56.22	0.69	92.90	59.27	1.46	94.04
	5×5	61.79	30.55	80.50	67.02	56.83	69.97	67.60	12.75	92.30	65.62	**0**	**97.68**	59.34	**0**	94.54	64.70	0.07	97.19
	7×7	60.42	61.93	64.39	63.56	16.22	88.81	58.16	18.74	84.85	61.24	0.10	95.71	58.39	**0**	94.34	61.42	0.20	**95.75**
Ftronjan	-	67.84	0.19	98.90	63.10	1.58	95.84	67.67	0.75	98.54	67.57	0.06	98.84	62.52	0.08	96.30	67.69	**0.03**	98.91
Blended	-	67.72	28.63	84.38	58.15	11.85	87.98	59.20	23.66	82.60	66.53	8.33	93.93	62.16	**5.98**	92.92	67.03	7.21	**94.74**
Input-aware	-	61.24	54.06	68.85	54.30	**1.80**	**91.51**	60.25	59.21	65.78	55.16	38.90	73.39	55.02	2.78	91.38	54.92	32.10	76.67
Avg	-	62.73	39.33	76.44	60.49	21.67	84.15	61.89	24.19	83.58	61.19	5.53	92.56	58.09	**1.29**	**93.14**	60.68	4.78	92.68

blending. IAB represents a dynamic backdoor attack, and Ftrojan is an invisible trigger backdoor attack based on frequency-domain Trojans. The experiments were conducted using the default attack configurations from BackdoorBench [36], with the poisoning rate set to 0.1.

Backdoor defense setting. We compare the proposed method with five SOTA backdoor defense methods that require access to benign training samples, namely Finetuning, Fine-pruning [10], Adversarial Neuron Pruning (ANP) [11], Neural Attention Distillation (NAD) [28], Implicit Backdoor Adversarial Unlearning (I-BAU) [12]. We also compare our backdoor defense framework with the method proposed in Backdoor Cleansing with Unlabeled Data (BCU) [15]. Notably, like our framework, BCU can only requires access to unlabeled out-of-distribution data. Tiny-ImageNet [37], GTSRB [38] are used as the out-of-distribution(OOD) datasets.

We conduct MSA for 10 epochs with a batch size of 128 and a learning rate of 0.01. Our MSA hyperparameters are set as $\gamma_i = 0.9$, $\beta = 0.1$. Based on [13] and extensive experiments, we found that a small β concentrates the trigger distribution, limiting its ability to capture the full distribution, while a large β disperses the output of G_i, reducing the average ASR. This configuration achieves the highest ASR and optimal trigger reconstruction.

We set the epochs for the proposed representation extraction process to 1 for backdoor knowledge and 5 for benign knowledge. The training epochs for backdoor unlearning are set to 15, using the Adam optimizer with a learning rate of 1×10^{-6} and a weight decay of 1×10^{-6}. For baseline methods, the training hyperparameters are adjusted based on BackdoorBench [36] and have access to 5% benign training samples.

Evaluation metric. In this work, three metrics are used to evaluate the quality of backdoor defense methods: clean ACCuracy (ACC), Attack Success Rate (ASR), and Defense Effectiveness Rating (DER) [39]. ACC represents the model's prediction accuracy on clean samples, while ASR measures the ratio of backdoor samples that are successfully classified into the target label. To comprehensively consider the changes in ACC and ASR and reflect the effectiveness of the defense method, we introduce DER, which is calculated using ACC

TABLE III: Defense results using different numbers of unlabeled out-of-distribution images.

Out-of-distribution dataset		Tiny (DER ↑)			GTSRB (DER ↑)			CIFAR-100 (DER ↑)		
Defense samples → ATTACK ↓		1000	2500	5000	1000	2500	5000	1000	2500	5000
Badnet	3×3	98.18	98.94	99.10	86.28	91.04	94.75	98.19	98.96	98.69
	5×5	98.24	98.46	98.68	91.02	93.15	95.59	98.68	98.81	98.98
	7×7	98.84	98.69	98.82	91.80	94.50	96.91	98.43	98.81	98.70
TrojanNN	3×3	98.30	99.00	97.67	89.74	94.68	94.66	98.78	99.08	98.94
	5×5	98.90	98.47	98.58	90.87	93.46	95.68	98.99	99.05	98.86
	7×7	98.72	98.86	98.91	91.79	94.29	96.95	98.50	98.80	98.30
Ftronjan	-	98.24	98.15	95.60	88.97	94.34	95.24	98.12	97.64	96.18
Blended	-	98.98	97.64	94.65	90.32	91.80	95.08	97.70	97.52	97.01
Input_aware	-	89.99	96.80	95.12	85.96	87.57	88.58	95.06	95.49	95.89
Avg	-	97.65	98.33	97.46	89.64	92.76	94.82	98.05	98.24	97.94

and ASR metrics based on the following formula:

$$DER = \frac{\max(0, \Delta ASR) - \max(0, \Delta ACC) + 1}{2} \tag{6}$$

B. Comparison with other defense methods

We compare our proposed framework with six state-of-the-art defense methods on CIFAR-10 and CIFAR-100 [34] in Table I and Table II, separately. For CIFAR-10, our defense method maintains an average ASR below 0.5% and achieves the highest average DER among almost all attacks compared to SOTA defenses, exceeding 98% when using Tiny-ImageNet and CIFAR-100 as OOD data. When CIFAR-100 is used as the in-distribution data, due to the increased complexity of the dataset, the challenge of defense also rises. However, our method still outperforms all baseline methods in terms of ASR and DER for the vast majority of backdoor attacks.

With the GTSRB dataset as OOD source, the average ASR reached the lowest at 0.19%. However, this also resulted in a significant impact on ACC, possibly due to the inconsistency between the standardized traffic sign images in the GTSRB dataset and the real-world object images predominantly present in the in-distribution dataset. To better evaluate the robustness of our method, we also tested the defense effectiveness when the trigger sizes are expanded to 5×5 and 7×7. The results indicate that our method is insensitive to trigger size and the defense performance improves as the trigger size increases.

FT, NAD, and ANP show generally higher ACC compared to other defenses, with FT achieving the highest average of 92.43%. However, they failed to significantly reduce the ASR of Ftrojan and Blended backdoor attacks to a reasonable level. The i-BAU method utilizes implicit hypergradient techniques to address min-max optimization, achieving an ASR of less than 3% against almost all five backdoor attacks, performing well compared to existing SOTA defense methods. However, this often comes at the expense of lower accuracies compared to other methods, resulting in an average ACC of only 85.86%. Although BCU achieves results comparable to defenses using in-distribution data against most SOTA attack methods, it faces challenges in eliminating backdoors while maintaining high ACC, leading our method to demonstrate a lower average ASR and a higher DER in comparison.

C. Analysis

Defense performance under different number of out-of-distribution samples. We utilized CIFAR-10 as the in-distribution dataset and Tiny-ImageNet, GTSRB, and CIFAR-100 as OOD datasets to evaluate the effectiveness of defenders with varying quantities of unlabeled OOD samples. For all three datasets, we randomly sample 1000, 2500, 5000 images. As shown in Table III, the defense effectiveness rating (DER) of our method remains consistent across different numbers of OOD samples, indicating that our defense method is not sensitive to the quantity of defense samples.

TABLE IV: Defense results under different poisoning ratio on CIFAR-10 using Tiny-ImageNet as out-of-distribution data.

Posion ratio			5%		10%		20%		40%	
ATTACK			No Defense	Ours	No Defense	Ours	No Defense	Ours	No Defense	Ours
Badnet	3×3	ACC ↑	92.24	85.15	91.60	90.15	90.83	87.78	88.01	85.15
		ASR ↓	100	0.45	99.98	0.63	100	0.24	100	0.78
	5×5	ACC ↑	92.90	89.62	93.41	90.46	92.17	88.95	90.73	86.11
		ASR ↓	100	0.36	100	0.13	100	0.22	100	0.54
	7×7	ACC ↑	92.94	89.83	93.37	90.83	91.98	88.12	91.57	88.19
		ASR ↓	100	0.18	100.00	0.08	100.00	0.10	100	0.43
TrojanNN	3×3	ACC ↑	91.86	85.30	91.87	90.28	90.80	87.58	88.53	85.93
		ASR ↓	99.98	0.28	99.75	0.17	100.00	0.56	100	0.50
	5×5	ACC ↑	92.56	90.06	93.28	90.34	92.55	90.55	91.36	88.30
		ASR ↓	100	0.37	100	0.12	100.00	0.22	100	0.41
	7×7	ACC ↑	92.81	90.12	93.43	91.31	91.98	88.62	91.27	87.30
		ASR ↓	100	0.81	100.00	0.16	100.00	0.10	100	0.31
Ftrojan	-	ACC ↑	92.85	88.03	93.32	90.54	92.00	89.28	90.82	87.41
		ASR ↓	99.98	0.28	100	0.92	100	0.09	100	0.15
Blended	-	ACC ↑	92.87	91.30	93.36	89.47	92.11	88.45	90.82	87.87
		ASR ↓	99.38	6.30	99.71	0.54	99.78	0.44	99.94	5.94
Input-aware	-	ACC ↑	90.53	88.21	91.32	88.94	88.41	87.49	88.41	88.05
		ASR ↓	89.93	0.30	96.85	0.86	99.54	2.50	95.86	2.75

TABLE V: Running time of different defense methods.

Running time(sec.)	FT	FP	i-BAU	NAD	ANP	BCU	Our
CIFAR-10	366.82	1166.12	335.29	294.29	**47.07**	414.27	<u>103.46</u>
CIFAR-100	307.12	1195.87	298.31	297.63	**55.13**	222.05	<u>136.60</u>

Defense Performance under Different Poisoning Ratios. To investigate the impact of poisoning ratios on our defense method against backdoor attacks, we conduct experiments using CIFAR-10 as in-distribution data and Tiny-ImageNet as OOD data. As shown in Table IV, the results indicate a slight decrease in the original ACC of the backdoor attack methods with increasing poisoning rates. Overall, our defense method demonstrates stable performance in most cases.

Running time comparison. All defense method experiments were conducted on one RTX 3090 GPU and PreAct-ResNet18 model. For BCU and our proposed defense, 2,500 Tiny-ImageNet images are employed as OOD data. The results are shown in Table V. Our method is faster than all other approaches except ANP on both datasets. Despite ANP's efficiency, our method achieves markedly better defense performance.

V. Conclusion

Inspired by the biological immune system, we propose a novel backdoor defense method using only unlabeled out-of-distribution data. We carefully design a defense framework, which first captures the valid trigger distribution via a max-entropy staircase approximator and then these recovered trigger patterns are leveraged to extract backdoor knowledge from the victim model through a knowledge distillation based method. Finally, we apply a machine unlearning technique optimized by KL-divergence maximization to thoroughly eliminate backdoor knowledge from the poisoned model. Extensive experiments validate that our method is effective on reducing backdoor threats against state-of-the-art backdoor attacks.

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